

Measured shape

Bending channels
Covariance + base pose

Observed environment

Plane points / normals
Covariance + friction

↓ All-time shape / environment likelihoods ↓

Previous state

Cosserat equilibrium
Latent contacts + tip

Current state

Cosserat equilibrium
Latent contacts + tip

↓ Friction: same material coordinate s_j ↓

Joint constrained MAP

Shape / environment likelihoods
Contact and friction complementarity
Initial + temporal state priors

↓
Contact arcs • force vectors • tip force

Physical audit • local uncertainty • ODE counts

Factor structure illustration